Contact & Data Volume

Da Vinci Satellite

Issue:	1
Revision:	0
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Author(s):	Nicolò Basso



VSV Leonardo Da Vinci

TU Delft

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Systems Team

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With the ESA Fly Your Satellite! 4 Programme

By

The Da Vinci Satellite Team

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Review and Approval

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Distribution

This document is meant to be distributed within the DVS team and with the ESA FYS Team.

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Introduction 1

This report estimates how much data the Da Vinci Satellite can exchange with the Delft ground station. The DVS Data Budget [1] derives the useful data rate from the TRXVU transceiver [2] and the AX.25 frame structure, but treats the contact time as an assumption (three passes of 10 min per day). Here the contact time is instead obtained from an orbit propagation, and combined with the data rates of the link budget [3] to give an estimate of the volume that can be transmitted on the downlink and on the uplink.

The work consists of three parts:

- a review of the existing ground-track scripts, with corrections to the visibility and pass-extraction logic;
- a one-year simulation of the passes over Delft, giving the statistical distribution of pass durations;
- the conversion of the simulated contact time into a data volume per pass and per day.

1.1 Assumptions

The following assumptions have been adopted for this work:

- A single ground station in Delft is used, and every pass above the elevation mask is scheduled and used.
- The baseline elevation mask is 10°, as assumed in the data budget [1]. The link budget closes with positive margin at this elevation (Section 5.1).
- Passes shorter than 2 min are not scheduled, as the time needed for acquisition and synchronisation leaves too little useful contact. This removes 44 of the 1806 passes above the mask. The data budget does not define such a threshold; it is an assumption of this report.
- The fraction of each pass usable for data transfer is $f_{\text{eff}} = 0.8$, as in the data budget, to account for acquisition, synchronisation and protocol handshaking.
- The transceiver is full duplex (separate VHF receiver and UHF transmitter), so uplink and downlink use the same contact time.
- Frames lost to bit errors are counted as lost data; the retransmission strategy is not modelled.
- No terrain or local obstructions are considered beyond the elevation mask.
- Drag uses the default TudatPy atmosphere, which corresponds to the US Standard Atmosphere 1976 (moderate solar activity). The real decay depends on the solar activity in 2028 and is the main uncertainty on the yearly statistics (Chapter 4).

Simulation Model 2

2.1 Orbit Propagation

The orbit is propagated with TudatPy [4] using the scripts in `simulations/`. The inputs are listed in Table 2.1. A full year is simulated, so that every phasing between the ground track, the station and the precessing orbital plane is sampled, giving 1762 passes.

Table 2.1: Simulation inputs.

Parameter	Value
Initial orbit	$a = 6890$ km, $e = 0.010$, $i = 52°$, $\omega = \Omega = \theta = 0$
Altitude over the year	426 – 579 km
Spacecraft	2.2 kg, $A = 0.0273$ m², $C_D = 2.2$, $C_R = 1.2$
Force model	Earth SH 5×5, drag, Sun and Moon point mass, SRP
Integrator	RK4, fixed step of 10 s
Simulated period	2028-01-01 to 2029-01-01 (366 days)
Ground station	Delft, 52.0° N, 4.4° E, on the WGS84 ellipsoid
Elevation mask	10° (baseline)

At every step the elevation of the satellite above the local horizon of the station is computed, and a pass is defined as an interval in which the elevation is above the mask. The start and end times of each pass are found by linear interpolation at the mask crossing. With a 10 s step the pass durations agree with a 2 s reference run to within 0.2 s, so the step size does not limit the accuracy.

2.2 Review of the Original Scripts

The original scripts were checked before producing the results. The propagation itself (environment, accelerations, initial state and integrator) was found to be sound. The following issues in the visibility and pass-extraction logic were corrected:

1. **Elevation mask.** Visibility was defined as a zenith angle of at most 75°, i.e. an elevation mask of 15°, while the data budget and link budget assume 10°. The mask is now an explicit parameter with a default of 10°.
2. **Station position.** The station was placed on a sphere of mean radius, with a geocentric vertical. At 52° latitude this places the station about 6 km too high and tilts the local vertical by about 0.19°. The station is now placed on the WGS84 ellipsoid [5] and the elevation is measured from the ellipsoid normal.
3. **Time resolution.** Pass start and end times were taken at sample points, with an error of up to one step at each edge. They are now interpolated at the mask crossing.
4. **Truncated passes.** A pass in progress at the start or end of the simulation window was counted as a short pass, which biased the mean duration low. Such passes are now discarded.
5. **Duplicated code.** The calendar script contained its own copy of the pass detection. Both scripts now use a shared function, which is checked against an analytic case in `simulations/test_passes.py`.

Pass Statistics 3

3.1 Pass Duration Distribution

Figure 3.1 shows the distribution of the duration of the 1762 usable passes above 10° (longer than 2 min), together with the fitted normal distribution. The satellite passes over Delft 4.81 times per day on average, for a mean of 6.69 min per pass and a total of 32.2 min of contact per day.

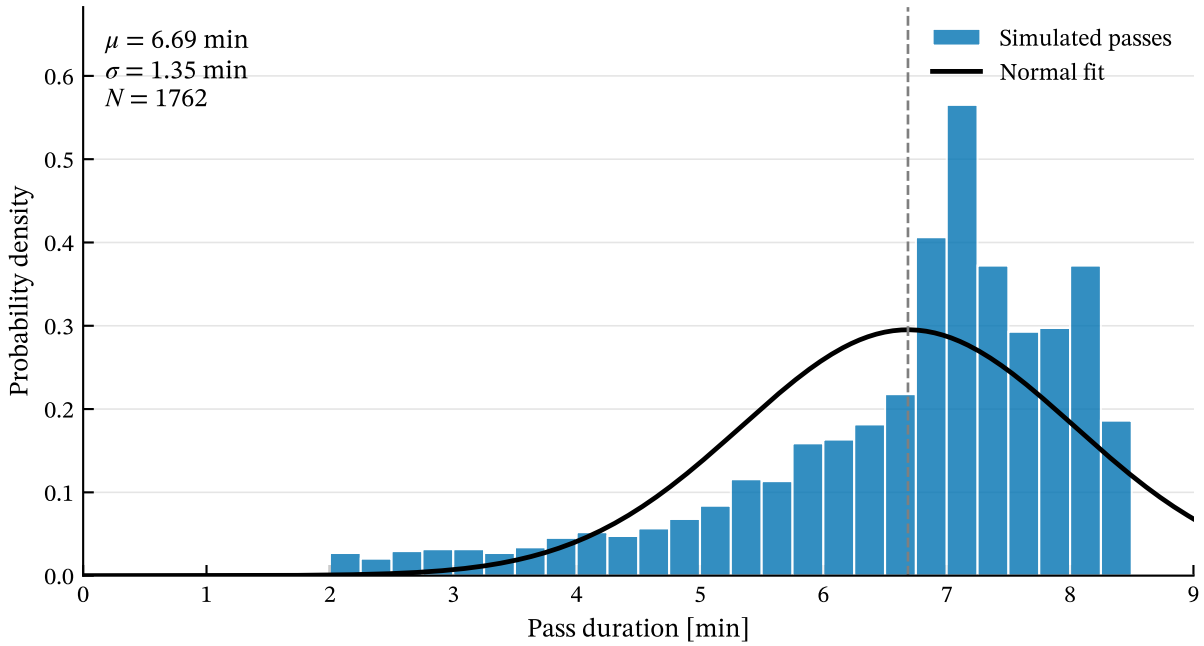


Figure 3.1: Distribution of pass durations above 10° elevation over one year, with the fitted normal distribution ($\mu = 6.69$ min, $\sigma = 1.35$ min).

The distribution is not exactly normal: it is bounded above by the duration of an overhead pass (8.48 min) and is skewed towards the long passes, with a tail of short, low-elevation passes. The normal fit is therefore a convenient summary, but the data volumes in Chapter 5 use the simulated durations directly. The median pass lasts 7.02 min; 10 % of the passes are shorter than 4.73 min and 10 % are longer than 8.11 min.

3.2 Daily Contact

Since data is generated and planned on a daily basis, the passes are also grouped per UTC day (by the start time of the pass). Figure 3.2 shows that every day has between 4 and 6 usable passes, and 4.81 on average ($\sigma = 0.52$). The resulting total contact time per day (Figure 3.3) has a mean of 32.2 min ($\sigma = 4.0$ min), ranging from 25.8 min to 44.9 min. On 90 % of the days the contact time is at least 26.6 min, which is used below as a “low day” for planning.

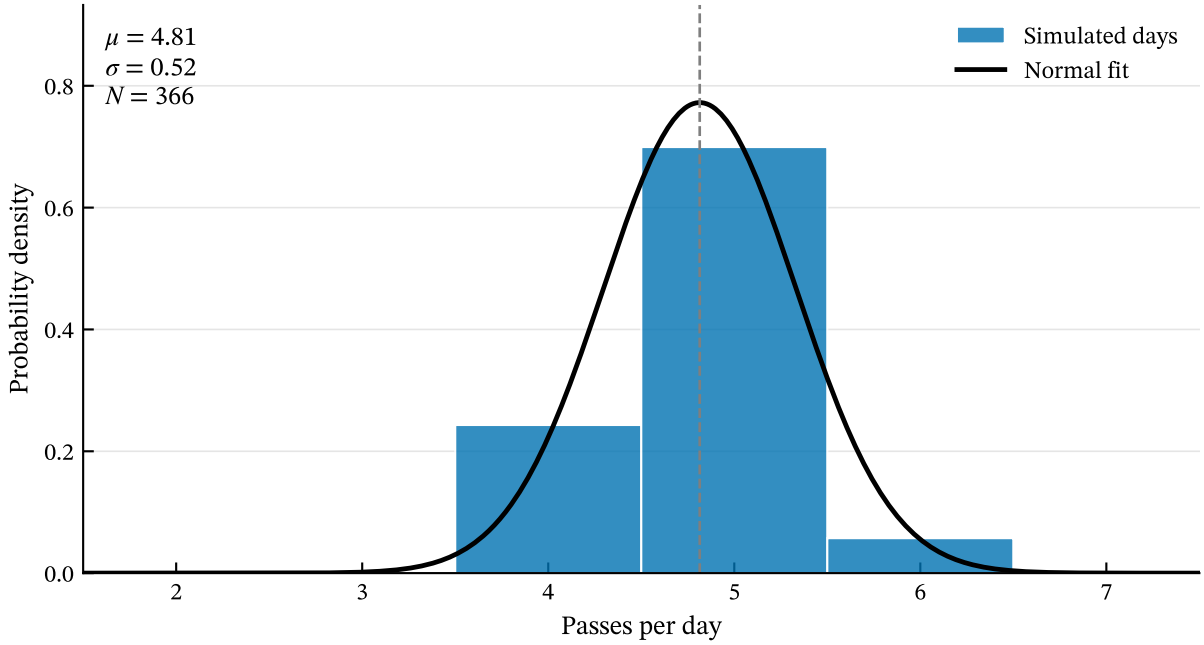


Figure 3.2: Distribution of the number of usable passes per day over one year, with the fitted normal distribution.

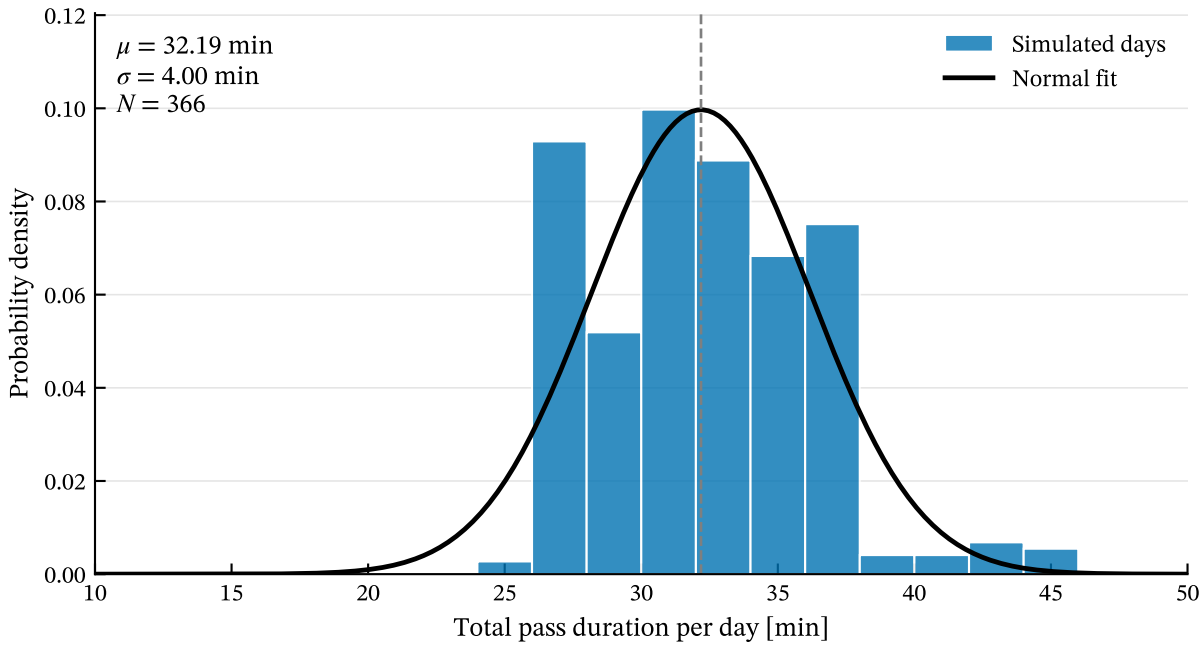


Figure 3.3: Distribution of the total pass duration per day over one year, with the fitted normal distribution.

The daily contact time is closer to a normal distribution than the single-pass duration, because it is the sum of several passes. It is nevertheless bimodal around 27 min and 33 min, corresponding to days with four and five passes, with a small tail of six-pass days above 40 min.

3.3 Effect of the Elevation Mask

Table 3.1 shows how the pass statistics change with the elevation mask. A lower mask gives both more passes and longer passes, so the daily contact time is very sensitive to the local horizon of the station.

Table 3.1: Pass statistics over one year for different elevation masks.

Mask	Passes	Per day	Mean [min]	σ [min]	P10 [min]	P90 [min]	Contact [min/day]
5°	1935	5.29	8.43	1.59	6.10	10.04	44.6
10°	1762	4.81	6.69	1.35	4.73	8.11	32.2
15°	1566	4.28	5.44	1.09	3.82	6.61	23.3
20°	1384	3.78	4.52	0.87	3.04	5.47	17.1

Verification 4

The pass prediction was verified in three independent ways, using `simulations/verify_passes.py`. The results are stored in `output/verification.json`.

4.1 Comparison with Heavens-Above (ISS)

To check that the passes are predicted correctly, they were compared with Heavens-Above [6], a widely used website that lists when satellites pass over a given location. DVS is not in orbit yet, so the International Space Station (ISS) was used instead: its orbit is very similar to the DVS one (51.6° inclination, about 420 km altitude). For Delft, Heavens-Above lists 47 ISS passes above 10° between 18 and 27 October 2026, each with the time at which the ISS rises above 10° and the time at which it sets below 10° again.

Heavens-Above computes the ISS position from its public orbit data (two-line elements [7]) with the standard SGP4 model [8]. The same orbit data was used here for two tests:

1. **Test A: pass detection only.** The ISS positions are computed with SGP4, as Heavens-Above does (using Skyfield [9]), but the passes are found with the code of this report. Any difference therefore comes only from how this work places the ground station, computes the elevation and finds the start and end of each pass.
2. **Test B: complete simulation.** The ISS is propagated with the same TudatPy model that is used for DVS, starting from the ISS position and velocity on 18 October. This tests the whole chain, including the orbit propagation.

For every pass, the predicted start time and duration were compared with the values listed by Heavens-Above. Table 4.1 gives the average and the largest difference.

Table 4.1: Difference between the predicted ISS passes and the 47 passes listed by Heavens-Above.

Test	**Start of the pass** average / largest [s]	**Duration of the pass** average / largest [s]
A: pass detection only	0.5 / 1.1	0.5 / 1.3
B: complete simulation, first 2 days	0.6 / 2.3	0.8 / 4.6
B: complete simulation, all 9 days	10.9 / 31.0	0.8 / 4.6

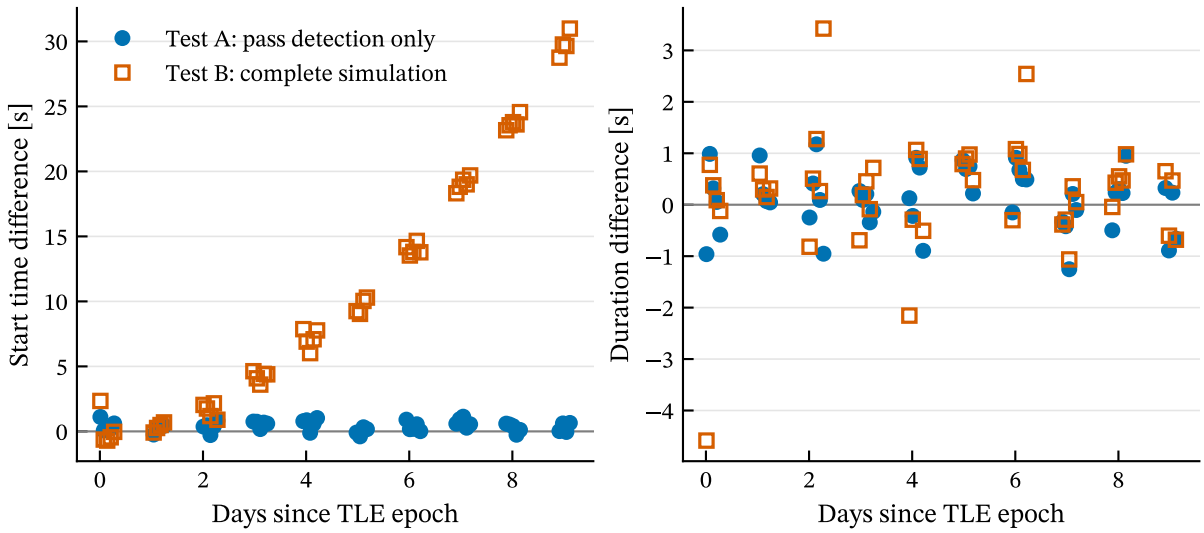


Figure 4.1: Difference with Heavens-Above for each ISS pass. Left: start time of the pass. Right: duration of the pass.

The results can be read as follows:

- **Test A** agrees with Heavens-Above to about one second, which is the precision of the published times (they are rounded to whole seconds). The ground station position, the elevation calculation and the pass detection are therefore correct.
- **Test B** agrees within 2 s during the first two days. After that, the predicted passes start slightly later every day, by about 3 s per day, reaching 31 s after nine days (left of Figure 4.1). This is expected: the two models use a different description of the atmosphere, so the simulated ISS is slowed down by drag at a slightly different rate and gradually falls behind. Such a drift is normal for any orbit prediction over several days.
- The **duration** of the passes, which is the only quantity used in the data budget, is not affected by this drift: it agrees within 0.8 s on average (right of Figure 4.1). The single larger difference of 5 s belongs to a pass that only just rises above the mask (maximum elevation 11°), where a tiny difference in the orbit has the largest effect on the duration.

In short, the simulation predicts how long the passes last as accurately as Heavens-Above, which validates the contact times used in this report.

4.2 Comparison with SGP4 over One Year (DVS)

The DVS orbit was also propagated over 2028 with SGP4, and the passes were found with Skyfield's own rise and set search, independent of the code of this work. The SGP4 mean elements were set equal to the TudatPy initial elements. Two cases were run: without drag, which isolates the geometry and gravity model, and with the SGP4 drag term computed from the DVS ballistic coefficient.

Table 4.2: Yearly pass statistics of the DVS orbit over Delft (10° mask, passes of at least 2 min).

Quantity	**TudatPy** (this work)	**SGP4** no drag	**SGP4** with drag
Passes per day	4.81	4.83	4.57
Mean pass duration [min]	6.69	6.90	5.96
Longest pass [min]	8.48	8.71	8.31
Contact per day [min]	32.2	33.3	27.2
Altitude at year end [km]	439 (perigee)	443 – 585	229 – 301

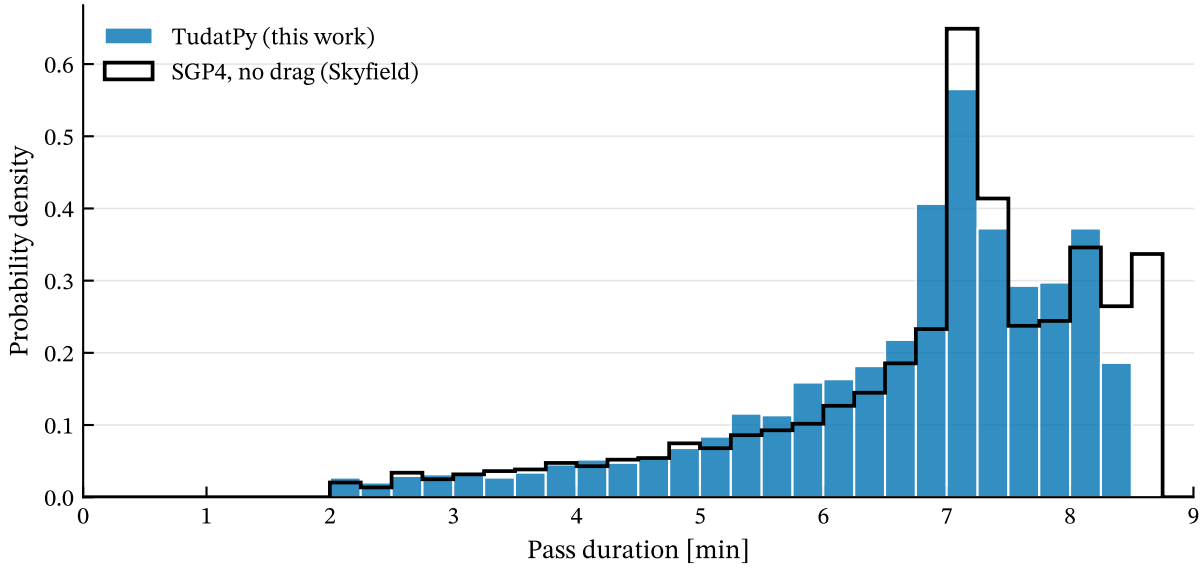


Figure 4.2: Pass duration distribution over 2028 from TudatPy and from SGP4 without drag.

Without drag, SGP4 gives the same number of passes per day to within 0.4 % and the same shape of the duration distribution (Figure 4.2). Its passes are 3 % longer, which is consistent with the slow decay included in the TudatPy run: the SGP4 orbit stays at its initial altitude. With the SGP4 drag term, on the other hand, the orbit decays to about 229 – 301 km within the year, and the contact time drops to 27 min per day. The SGP4 density profile is fixed and not intended for long-term decay prediction, so this case is a pessimistic bound rather than a prediction. It shows, however, that the orbit decay is the dominant uncertainty on the yearly statistics: the geometry and pass extraction are verified, but the altitude at which the satellite will fly is not. A lifetime analysis with a solar-activity-dependent atmosphere (e.g. NRLMSISE-00 with predicted solar flux) is recommended to refine the yearly figures.

4.3 Analytical Maximum Pass Duration

For a circular orbit at altitude h, the longest pass, directly overhead, lasts [10]

$$T_{\max} = \frac{P}{\pi}\left(\frac{\pi}{2} - \varepsilon - \eta\right), \quad \sin\eta = \frac{R_E \cos\varepsilon}{R_E + h} \tag{1.}$$

where P is the orbital period and ε is the elevation mask. At the highest altitude reached during the year (579 km) this gives 8.27 min, and at the lowest (426 km) 6.52 min. Eq. 1 neglects the rotation

of the Earth. Since the station latitude is close to the inclination, the longest passes occur near the northern turning point of the orbit, where the satellite moves eastwards together with the station. This reduces the relative angular rate by a factor $1 - (\omega_E / n)\cos i$ and lengthens the pass up to 8.63 min. The longest simulated pass, 8.48 min, lies between these two bounds.

Data Volume 5

5.1 Link Budget Check

The link budget [3] is computed at 500 km altitude for elevations from 5° to 90°. The UHF downlink at 9600 bit/s has a link margin of +3.3 dB at 5° and +5.2 dB at 10°, after a 3 dB system margin and a required E_b/N_0 of 10.4 dB. The VHF uplink at 2400 bit/s has a margin of more than 30 dB. The link therefore closes over the whole pass above the 10° mask, and the full data rate can be used for the entire contact.

5.2 Method

Following the data budget [1], the useful volume transferred during a pass of duration T_{pass} is

$$V_{\text{pass}} = R_{\text{phys}} \cdot \eta \cdot (1 - \text{PER}) \cdot T_{\text{pass}} \cdot f_{\text{eff}} \tag{1.}$$

where R_{phys} is the data rate from the link budget, η is the payload efficiency of the frame type (payload bytes over total frame bytes, including the AX.25 metadata and the frame header) and $\text{PER} = 1 - (1 - \text{BER})^L$ is the packet error rate for a frame of L bits at $\text{BER} = 10^{-5}$. The daily volume is obtained by replacing T_{pass} with the simulated contact time per day, both for the mean day and for a low day (10th percentile of the daily contact time). The scenarios and frame sizes are taken from the data budget: for the downlink, telemetry frames (30 B payload in 60 B), picture chunks (64 B in 88 B) and a mixed case with 25 % telemetry and 75 % picture chunks by time; for the uplink, code or picture chunks (64 B in 95 B) and full-size frames (193 B in 218 B). As in the data budget, 1 KB = 1024 B and 1 MB = 1024 KB.

5.3 Downlink

Table 5.1: Downlink data volume (UHF, 9600 bit/s, 10° mask).

Scenario	**Useful rate** [B/s]	**Short pass** P10 [KB]	**Mean pass** [KB]	**Long pass** P90 [KB]	**Low day** P10 [MB]	**Mean day** [MB]
Telemetry only	597	132	187	227	0.73	**0.88**
Payload only	867	192	272	329	1.06	**1.28**
Mixed 25/75	799	177	250	304	0.97	**1.18**

In the nominal mixed scenario the satellite can downlink about **250 KB in an average pass** and **1.18 MB per day** (Table 5.1). Short passes (P10) still carry 177 KB, which is enough for a full telemetry dump.

5.4 Uplink

Table 5.2: Uplink data volume (VHF, 2400 bit/s, 10° mask).

Scenario	**Useful rate** [B/s]	**Short pass** P10 [KB]	**Mean pass** [KB]	**Long pass** P90 [KB]	**Low day** P10 [MB]	**Mean day** [MB]
Code / picture chunks	201	44	63	76	0.24	**0.30**
Full-size frames	261	58	82	99	0.32	**0.38**

When uploading code or pictures in 64-byte chunks, about **63 KB per average pass** and **0.30 MB per day** can be uplinked (Table 5.2). The uplink is limited by its lower data rate rather than by the link margin.

5.5 Comparison with the Data Budget

- **Contact time.** The data budget assumed 3×10 min $= 30$ min per day. The simulation gives 4.81 passes of 6.69 min on average, i.e. 32.2 min per day at a 10° mask. The data budget has no minimum pass duration; the 2 min threshold used here removes only 2.4 % of the passes and has a negligible effect on the daily contact time.
- **Uplink data rate.** The data budget uses 9600 bit/s for the uplink, while the link budget is computed at 2400 bit/s. This report follows the link budget. At 9600 bit/s the uplink volumes would be four times higher, and the link would still close with about 25 dB margin.
- **Packet error rate.** The data budget quotes a PER of about 1.6 % for a 235-byte frame, while $1 - \left(1 - 10^{-5}\right)^{1880} = 1.86$ %. Frame losses are included here and reduce the volumes by 0.6 % to 2 %.

Conclusion

6

A one-year propagation of the Da Vinci Satellite orbit gives 4.81 passes per day over Delft above 10° elevation, with a mean duration of 6.69 min ($\sigma = 1.35$ min) and 32.2 min of contact per day. With the data rates of the link budget and the frame overheads of the data budget, the expected useful data volume is:

Table 6.1: Expected data volume for nominal operations.

Link	**Scenario**	**Per pass** [KB]	**Per day** [MB]
Downlink	Mixed 25/75	250	**1.18**
Uplink	Code / picture chunks	63	**0.30**

The pass prediction was verified against the published ISS passes over Delft (agreement within about 1 s on the geometry), against an independent SGP4 propagation over one year (within 3.5 % on the daily contact time when drag is excluded), and against the analytical maximum pass duration. These values replace the assumed contact parameters of the data budget. The main remaining uncertainties are the orbit decay over the mission, the usable fraction of each pass (f_{eff}), the actual horizon mask of the Delft station, and the uplink data rate, which should be aligned between the data budget and the link budget.

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